

REQUEST FOR QUOTE

AI-ENABLED DIGITAL PUBLISHING AND E-COMMERCE PLATFORM – PRODUCT DEVELOPMENT PROPOSAL

Note to Technology Partner: Please review the requirements in this document and add, modify, or remove items as necessary. Kindly include all mandatory information required at this stage to proceed with the proposal.

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INTRODUCTION

The organisation was established in the early 1960s as a bookshop and began publishing Islamic books in Malay and Arabic in 1963. By the 1980s, its publishing portfolio had expanded to include children's fiction, general readership novels, and madrasah textbooks. Over the decades, it has collaborated with notable authors who have contributed significantly to the development of Malay literature, cultural identity, and sociopolitical awareness in Singapore.

Today, the organisation focuses on producing children's books with strong Asian and local cultural contexts, delivered across multiple formats including print, e-books, and audiobooks. In parallel, it continues its translation and distribution activities and actively advocates for the sustainability and growth of Malay-language publishing in Singapore.

OBJECTIVES

The objectives of this grant proposal are as follows:

- Provide targeted financial support to help SMEs develop and implement online retail platforms.
- Enable the adoption of e-commerce technologies to strengthen SME competitiveness.
- Deliver training and capacity-building programmes to support effective online operations.
- Promote a culture of digital entrepreneurship and innovation among SMEs.
- Drive economic growth by expanding SME market reach, increasing revenues, and strengthening Singapore's digital economy.
- Embed AI-powered automation and analytics to enable intelligent content discovery, predictive user engagement, and data-driven decision-making.
- Leverage AI to enhance platform scalability, operational efficiency, and long-term sustainability for SMEs.

PROJECT SCOPE

The project will involve the following key activities:

- 1. Requirements Analysis**
Identify stakeholder needs and define functional and technical specifications.
- 2. UI/UX Design and Development**
Create wireframes, prototypes, and a modern, intuitive interface to enhance usability.
- 3. User Management System**
Develop secure authentication, registration, and personalised user settings.
- 4. Security and Data Protection**
Apply robust encryption and ensure compliance with data protection regulations.
- 5. E-book Format Conversion**
Integrate tools to support multi-format conversion and cross-device compatibility.

6. **Automated Content Tagging & Metadata Optimisation (AI)**
Use AI-generated categorisation to improve discoverability.
7. **Smart Search & Natural Language Processing (NLP)**
Improve search accuracy through semantic understanding of queries.
8. **AI-Powered Recommendation Engine**
Deliver personalised content suggestions based on user behaviour and preferences.
9. **Predictive Analytics & User Behaviour Modelling**
Anticipate user needs, optimise content placement, and forecast demand trends.
10. **Interactive Features**
Implement highlighting, bookmarking, dictionary lookup, and note-taking to improve engagement.
11. **Social Media Integration**
Enable content sharing to increase platform visibility and user reach.
12. **Fraud Detection & Anomaly Monitoring (AI)**
Enable real-time detection of suspicious transactions and access patterns.
13. **Testing and Quality Assurance**
Conduct functional and usability testing to ensure reliability and performance.
14. **Deployment and Monitoring**
Deploy the platform to a production environment and monitor post-launch performance.

Online Application / App Platform Features:

1. For Readers

Home Page

- Latest Titles (Book Categories)
- Featured Books
- Best-Sellers
- Showcase (Fast-Moving Books)
- Free Books
- Authors
- Featured Authors
- International Titles (Phase 3)
- Search books by name and author
- Banner for new launches
- AI-Based Personalised Home Page (dynamic content based on preferences and behaviour)
- Smart Content Discovery (AI-curated recommendations)
- Voice-Based Search (natural language queries)

Books Menu

- List all categories
- Search and Sort
- Add to Cart
- AI-Suggested Categories & Filters (based on reading behaviour)

Free Books

- Curated free titles
- AI-Curated Free Book Suggestions

Special Offers

- Promotions and discounts
- AI-Personalised Offers

Reader Online Platform Information

Authors Menu

- List of all authors with profile links
- Search authors by name
- AI-Suggested Authors based on reading patterns

Conference / Events

- List of conferences with blog links
- List of events with event links
- AI-Based Event Recommendations

Newsletter Subscription

- Subscription form

AI-Driven Content Personalisation

Contact Us

Social Media Links

- Facebook
- Twitter
- Instagram
- AI-Optimised Social Sharing Recommendations

Cart and Billing

- Add to cart
- Edit, delete, update cart
- Checkout page
- Billing

- Download after payment
- Email with download link
- AI-Based Upsell & Cross-Sell Suggestions

User Login Features

- Sign up (username, password, social login)
- Edit/Update profile
- Forgot/Reset password
- View my cart
- View my books
- View my wishlist
- AI-Based Reading History & Preference Tracking

2. For Authors & Publishers

- AI-Driven Pricing Recommendations (based on demand, genre trends, and competition)
- Predictive Sales Forecasting (future demand and revenue trends)
- Smart Keyword & Metadata Suggestions (to improve discoverability)
- AI-based market trend analysis
- AI-powered content performance dashboards
- Automated content tagging and categorisation

3. For Admin

- AI Dashboards displaying:
 - Churn risk
 - Sales velocity
 - User behaviour patterns
- AI-powered fraud detection & anomaly monitoring
- AI-based content moderation
- Predictive demand and inventory analytics
- Automated system alerts and insights

User Features:

- User can sign up by submitting email Id.
- User can sign up using social login (Facebook, Google, LinkedIn)
- Email verification
- User can Edit/Update Profile
- User can reset the password through forgot password
- User can view the list of books (hardcopy & digital books)
- User can view the purchased by him/her
- User can purchase a book
- User can do payment through different payment methods

- can check the status of the purchase
- User can go to help to contact the admin
- User can add books to his/her wish list
- User can subscribe to the latest news
- User can see the newly arrived books
- User can search based on author, publisher and book name
- User can see the categories of the books
- AI-curated book feeds, reading recommendations, and interest clustering

Publisher Features:

- Publisher can onboard to the platform by the submitting all the mandatory information.
- Publisher can login through email Id and password
- Publisher can create listing and list all the books that are to be sold
- Publisher can add/remove book in the listing
- Publisher can update books information in their listing
- Publisher can see the list of books sold and other statistics through a dashboard
- Demand forecasting, market gap analysis, content trend predictions

Author Features:

- Author can onboard to the platform by the submitting all the mandatory information.
- Author can login through email Id and password
- Author can create listing and list all the books that are to be sold
- Author can add/remove book in the listing
- Author can update books information in their listing
- Author can see the list of books sold and other statistics through a dashboard
- AI insights on best-performing genres, price optimisation, and reader behaviour

Admin Features

- List all users
- Add/Remove users
- Add/Remove authors
- Add/Remove publishers
- Approve/Reject authors and publishers
- Add/Remove books
- Add/Remove categories
- View sales and other reports
- View all Subscribers
- AI-based fraud alerts, automated moderation, anomaly detection

User Categories

The software is proposed to be used by different categories of users namely:

- Reader
- Author
- Publisher (Phase 2)
- International Publishers and Partners (Phase 3)

USE CASES

- Account Creation (Online Application / App)
- AI-Based User Segmentation
- Real-Time Personalisation
- Intelligent Content Discovery
- Browsing and Selection
- Shopping Cart
- Dynamic Pricing Optimisation
- Checkout Process
- Payment Confirmation
- Download/Access
- Account Management
- Automated Metadata Generation
- Customer Support
- Automated Customer Support (Chatbots)
- Predictive Sales Forecasting

Use Case of Publishing E-Books:

- Create the E-book
- Editing and Proofreading
- Choose a Platform
- Create an Account
- Upload Your E-book
- Set Pricing and Royalties
- Preview and Publish
- Marketing and Promotion
- Monitor Sales and Analytics
- Handle Customer Service
- Update and Promote
- Payment and Payouts

Core Technical Infrastructure Features

1. Digital Rights Management (DRM)

Integrated DRM capabilities will be provided to protect author and publisher content, supported by partnerships with established ePub and metadata specialists.

2. Large-Scale Data Handling

The platform will support high-volume data processing, including animated and interactive formats, through optimised and scalable technologies.

3. Data Protection

Sensitive information will be safeguarded using industry-standard encryption and secure coding practices to mitigate cyber vulnerabilities.

4. Secure Transactions

Critical data, including payment and personally identifiable information (PII), will be protected through tokenisation and data-masking techniques.

5. Authentication and Access Control

Strong authentication mechanisms, including multi-factor authentication (MFA), will be implemented alongside role-based (RBAC) and attribute-based (ABAC) access controls.

6. Secure Communication

All data transmission will be protected using secure protocols (TLS/SSL), with secured APIs employing OAuth, tokens, or API keys and input validation safeguards.

7. Input Validation and Threat Prevention

User inputs will be validated and sanitised to prevent vulnerabilities such as SQL injection, cross-site scripting (XSS), and other injection attacks.

8. Secure Development Practices

The platform will adhere to OWASP standards, supported by regular code reviews, vulnerability assessments, and security audits.

9. AI & Analytics Infrastructure

- Real-time data pipelines
- Machine learning models for recommendations
- Behavioural analytics engines
- AI-driven dashboards
- Automated insights generation
- Event-driven analytics

10. Key Differentiators and User Enablement Strategy

- Ease of Enrolment and Usage
- User-Centric Platform Design
- Systematic Content Categorisation
- Comprehensive Value Proposition

Customer Experience and Driven Audience Profiles

<i>Persona</i>	<i>Core Needs</i>	<i>Key Motivations</i>	<i>Attractive Platform Features</i>	<i>Value Delivered</i>
<i>Gen Z Digital Natives</i>	Speed, interactivity, social connection	Entertainment, self-expression, discovery	AI-based recommendations, interactive e-books, gamification, social sharing	High engagement, repeat usage, viral growth
<i>Busy Professionals</i>	Convenience, flexibility, time efficiency	Productivity, on-the-go access	Offline reading, cross-device sync, quick search, bookmarks	Increased daily usage, retention
<i>Socially Conscious Readers</i>	Purpose-driven content, inclusivity	Social impact, ethical consumption	Curated diverse collections, eco-friendly formats, cause partnerships	Brand loyalty, advocacy
<i>Lifelong Learners</i>	Knowledge, skill-building, structure	Self-improvement, career growth	Educational libraries, note-taking, learning integrations, AI summaries	Long-term engagement, subscription stickiness
<i>Audiobook Enthusiasts</i>	Hands-free consumption, convenience	Multitasking, accessibility	Large audiobook library, smart playback, listening-based recommendations	Extended session duration, accessibility inclusion

Technology development for E-Book

User Experience Optimisation

- Seamless and Intelligent Experience: Optimise platform speed, usability, and navigation, supported by AI-driven personalisation and intuitive content discovery.
- Enhanced Reading Functionality: Enable bookmarking, note-taking, highlighting, and cross-device synchronisation for uninterrupted and enriched engagement.
- Sustainable Business Value: Improved user experience drives higher satisfaction, stronger retention, increased referrals, and recurring revenue through repeat usage and subscriptions.

Content Expansion and Strategic Partnerships

- Diverse Content Growth: Collaborate with publishers, authors, and content providers to expand the catalogue across multiple genres and formats.
- Exclusive Offerings: Secure partnerships for exclusive titles and early-access releases to differentiate the platform.
- Personalised Discovery: Implement an AI-driven recommendation engine to enhance relevance and user engagement.

Monetisation Strategies

- Flexible Revenue Models: Implement subscription plans, pay-per-book options, or hybrid models to meet diverse user preferences.
- Tiered Pricing Structure: Offer multi-level plans with differentiated access, features, and benefits to maximise value capture.
- Brand Partnerships and Advertising: Enable targeted advertising, sponsorships, and relevant brand collaborations to create additional revenue streams.

Data Analytics and Customer Insights

- Behavioural Intelligence: Deploy robust analytics to capture user behaviour, preferences, and reading patterns.
- Personalised Growth Levers: Use insights to refine recommendations, personalise offers, and enable upselling and cross-selling.
- Continuous Improvement: Integrate feedback mechanisms to iteratively enhance platform features and usability.

AI-Enhanced User Experience Optimisation

- AI-driven personalisation, smart navigation, and predictive discovery
- Behaviour-aware reading tools and adaptive learning interfaces
- Cross-device intelligence with session continuity

Advanced Data Intelligence Layer

- Real-time behavioural analytics
- Predictive segmentation and churn modelling
- Automated content optimisation

Current Business Situation

Key Market Gaps

The current e-commerce market is dominated by Lazada and Shopee, which impose strict fulfilment requirements, delivery penalties, and increasing commission charges. Operational challenges such as failed pickups by assigned logistics partners further impact suppliers. Notably, these platforms do not support digital content sales.

By launching its own e-commerce platform, Customer aims to become a one-stop hub for Malay-language publications, covering both print and digital formats. This will enable Customer to:

- Control fulfilment and logistics
- Reduce platform commission costs
- Introduce exclusive digital content in Singapore
- Integrate social media campaigns and loyalty programmes to drive sustained sales

Phases



Development Approach

- The project will follow an Agile methodology, ensuring flexibility and continuous improvement.

- Development will be executed in 2–4 week sprints
- Regular stakeholder engagement and user feedback will guide iterative enhancements

Technology Partner Overview

This project aims to develop a modern, secure, and user-centric web platform for distributing both hardcopy and e-books. Built using advanced web technologies, the platform will deliver a seamless reading experience across devices.

JETS is uniquely positioned to deliver this solution, with proven expertise in next-generation software development and e-commerce platforms.

Project Team

- Project Manager: Overall coordination and delivery
- Front-End Developers: UI and interactivity
- Back-End Developers: Server, APIs, and databases
- UX/UI Designer: User experience and visual design
- QA Specialist: Testing and performance assurance
- Technical Writer: Documentation and user guides

Quality Assurance

- Comprehensive functional and edge-case testing
- Automated testing for continuous integration
- Structured bug tracking and resolution
- User Acceptance Testing (UAT) with real users

Cybersecurity Framework for Web Projects

1. Risk Assessment
2. Data Encryption
3. Secure Coding Practices
4. Authentication & Authorisation
5. Session Management
6. Firewalls & Intrusion Detection
7. Security Audits & Testing
8. File Upload Security
9. Content Security Policy (CSP)
10. Security Headers
11. Regular Backups
12. Incident Response Plan
13. User Education
14. Regulatory Compliance
15. Monitoring & Logging
16. AI-Driven Security

Digital Rights Management (DRM) Overview

Digital Rights Management (DRM) refers to a set of technologies and policies designed to protect digital content and intellectual property by controlling access, usage, and distribution. It is widely applied to digital assets such as e-books, software, music, films, and multimedia files. The primary objective of DRM is to safeguard the rights of content creators, publishers, and distributors while enabling secure and lawful content consumption.

Key Components of DRM**1. Access Control**

- DRM restricts content access to authorised users or devices through defined control mechanisms.

2. Copy Protection

- Technologies such as encryption and watermarking are used to prevent unauthorised duplication and redistribution.

3. Licence Management

- Usage terms are governed by digital licences that define how content may be accessed, shared, or stored.

4. Encryption

- Content is encrypted to ensure that only authorised users with valid decryption credentials can access it, protecting data during transmission and storage.

5. Authentication and Authorisation

- DRM systems verify user identity and enforce access rights based on licensing conditions.

6. Digital Watermarking

- Invisible identifiers are embedded within content to trace origin and detect unauthorised distribution.

7. Device Restrictions

- Usage may be limited to specific devices or a defined number of installations to prevent uncontrolled replication.

8. Platform Compatibility

- DRM solutions are designed to function across multiple platforms while maintaining content security through standardised protocols.

9. AI-Supported DRM Enhancements

- AI-based piracy detection
- Behavioural anomaly tracking
- Smart watermark tracing
- Automated licence violation alerts

RFQ - REQUIREMENTS AND COMMERCIALS

Section 1: Commercial & Pricing Details

S. No.	Requirements Category	Details to be Provided by Technology Partner	Pricing in INR
1.	Development Cost with Cost Breakdown for the following:	One Time	
A	• Modules		
B	• Features		
C	• Phases		
2.	Itemised Cost	Usage Based	
A	• Licensing		
B	• API		
C	• Cloud Hosting		
D	• Support		
3.	Payment Terms & Applicable Taxes	MIle-stones / Monthly / quarterly / Yearly	
4.	Validity of Quote		
5.	Optional Cost		
6.	Cost Escalation		

Section 2: Project Timeline & Schedule

T. No.	Time Line	Technology Partner Input	Remark
1.	Project Duration		
A	• Phase-wise (Timeline) (Analyse / Design / Development / Testing / Deployment)		
B	• Milestones Schedule (Key Deliverable with dates)		

C	Review / Approval Cycles (Number of Review Iterations)		
D	Go-Live Timeline		
E	Dependencies (Client Side Third Party Dependencies)		

Section 3: Technical Architecture and Information

U. No.	Time Line	Technology Partner Input	Remark
1.	Solution Overview	High Level Technical Architecture	
2.	Technology Stack	Programming language, frameworks, tools	
3.	Deployment Model	Cloud / On-premise / Hybrid	
4.	Hosting Details	Cloud provider, region, scalability	
5.	Database Technology	Type, version, scalability approach	
6.	Security Standards	Encryption, access control, compliance	
7.	Compliance	ISO, SOC, PDPA, GDPR (if applicable)	
8.	Performance Metrics	Expected response time, throughput	

Note: Information pertaining to internal and external integration's, support and maintenance services, service level agreements (SLAs), and the resource and governance model will be provided upon confirmation.